

Root [0]

Technical Sheet Only!
Please refer to the subsequent sheets for schematic

Power



File: power.kicad_sch

FPGA_Power



File: fpga_power.kicad_sch

FPGA_Config



File: fpga_config.kicad_sch

JTAG



File: jtag.kicad_sch

Serial



File: serial.kicad_sch

USB



File: usb.kicad_sch

Connectivity



File: connectivity.kicad_sch

Reviewer: Michal Varsanyi
Designer: Michal Varsanyi
Czech Technical University in Prague

Sheet: /
File: Controller_Board.kicad_sch

Title: Root

Size: A4 Date: 2026-03-10

KiCad E.D.A. 9.0.6

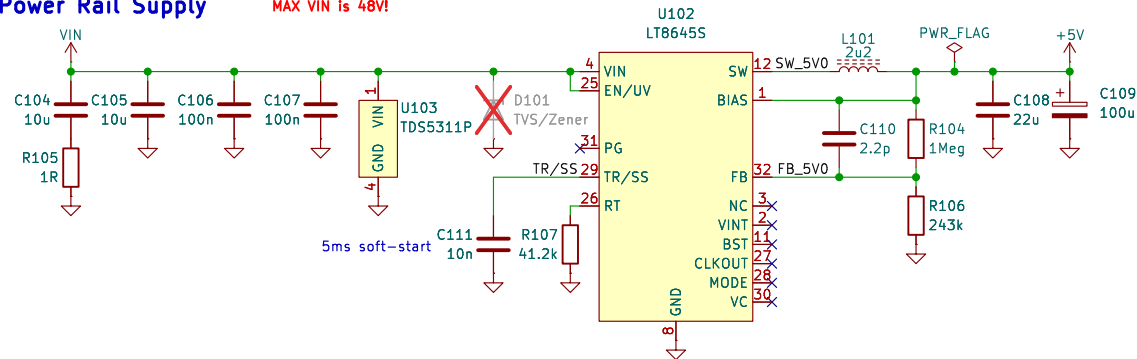
Rev: 1.0

Id: 0/8

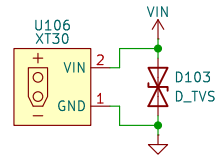
Power [1]

Power Rail Supply

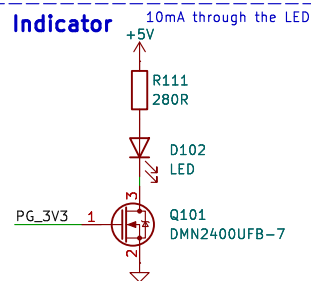
MAX VIN is 48V!



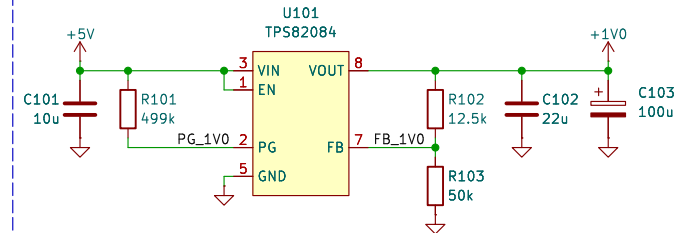
Input Connector



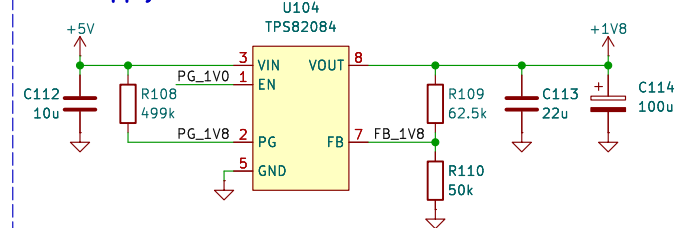
PG Indicator



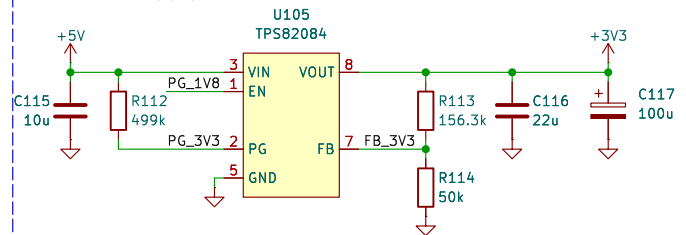
Core Supply



AUX Supply



IO Bank Supply



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Sheet: /Power/
File: power.kicad_sch

Title: Power

Size: A4 Date: 2026-03-07

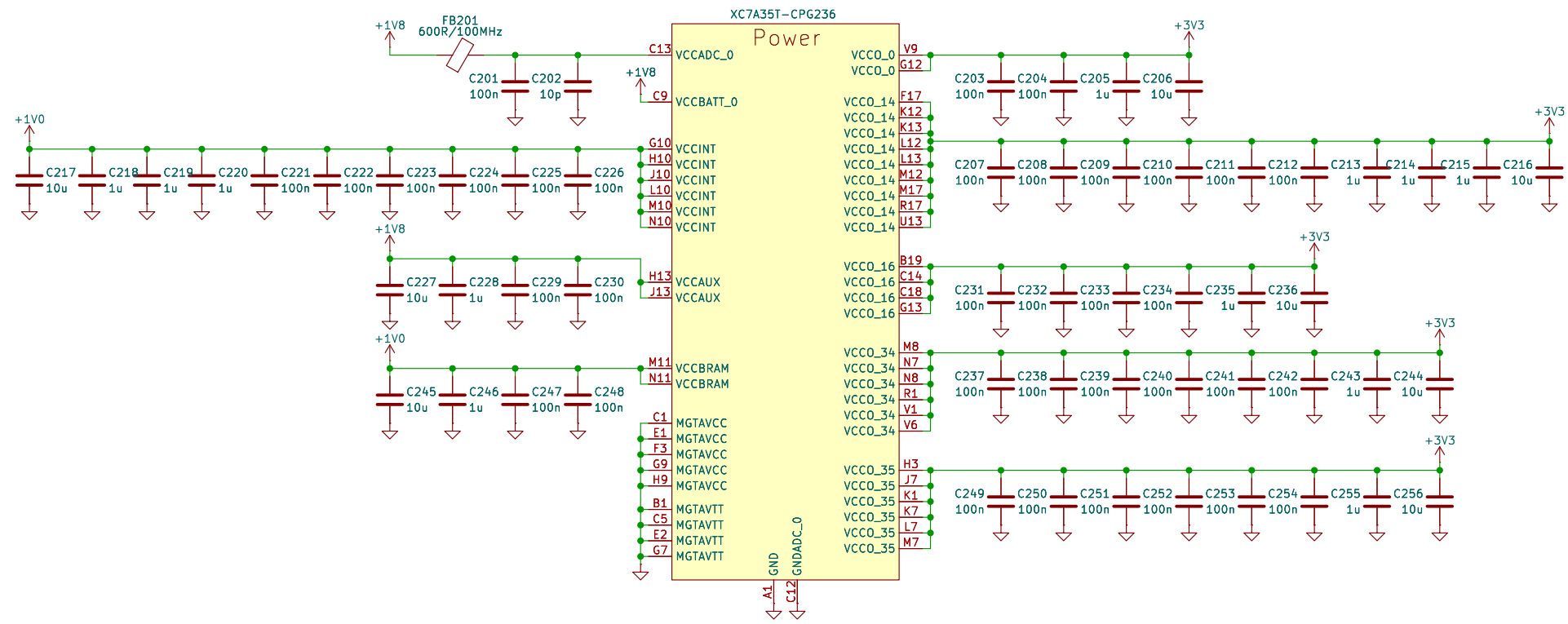
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Rev: 1.0

Id: 1/8

FPGA Power [2]

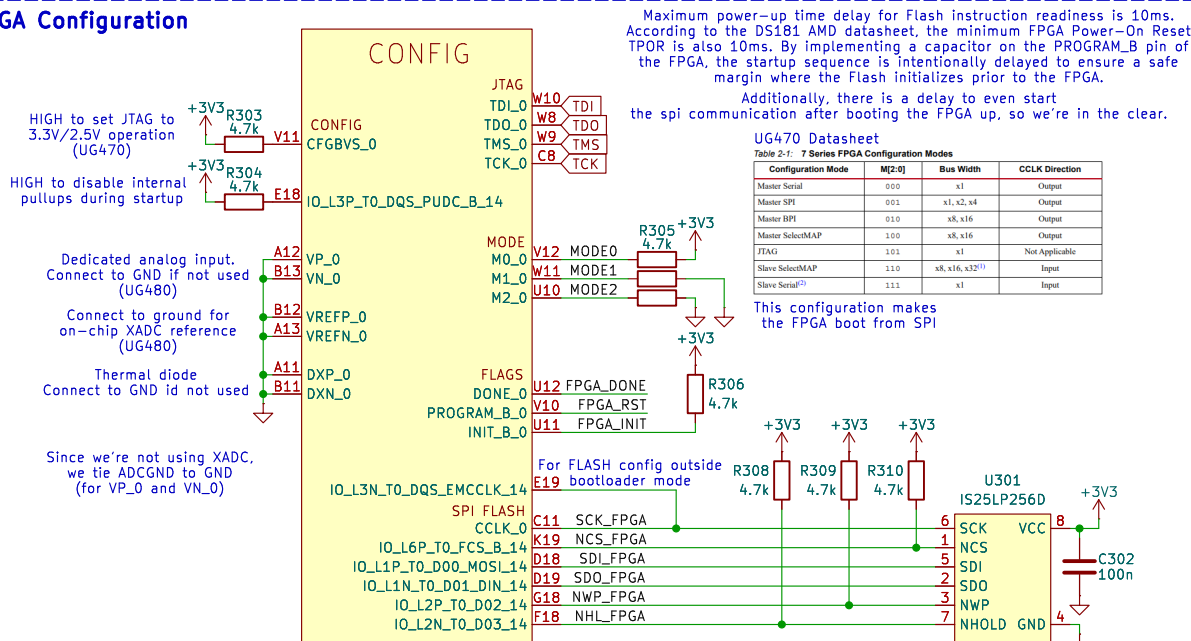
FPGA Power



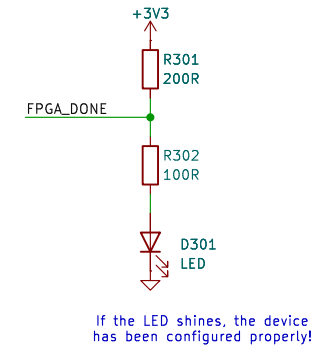
Reviewer: Michal Varsanyi
Designer: Michal Varsanyi
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Sheet: /FPGA_Power/
File: fpga_power.kicad_sch
Title: FPGA Power
Size: A4 | Date: 2026-03-10
KiCad E.D.A. 9.0.6 | Rev: 1.0
Id: 2/8

FPGA Config [3]

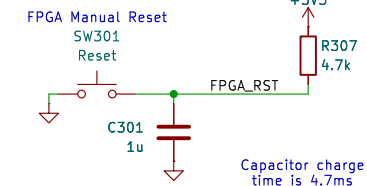
FPGA Configuration



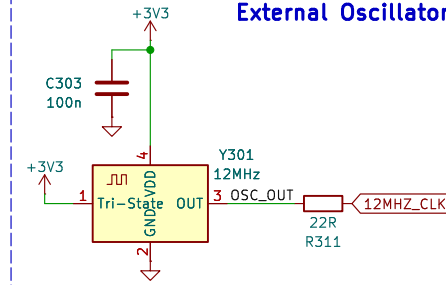
Configuration DONE



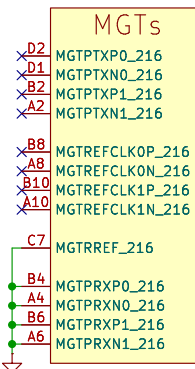
FPGA Manual Reset



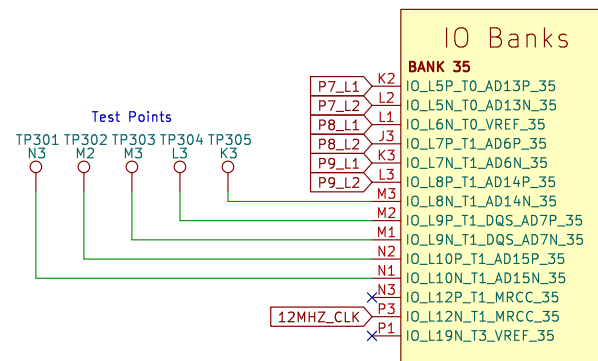
External Oscillator



Multi-Gigabit Transceivers



FPGA IOs



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Sheet: /FPGA_Config/
File: fpga_config.kicad_sch

Title: FPGA Config

Size: A4
Date: 2026-03-10
KiCad E.D.A. 9.0.6

Rev: 1.0
Id: 3/8

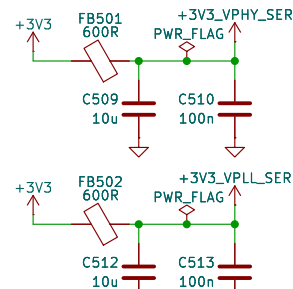
D

Id: 4/8

Serial [5]

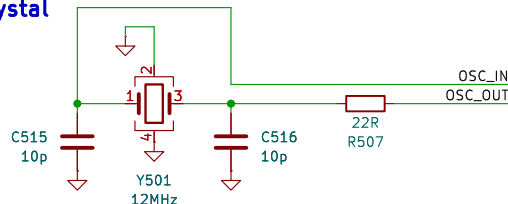
FTDI USB-to-UART

Filtering

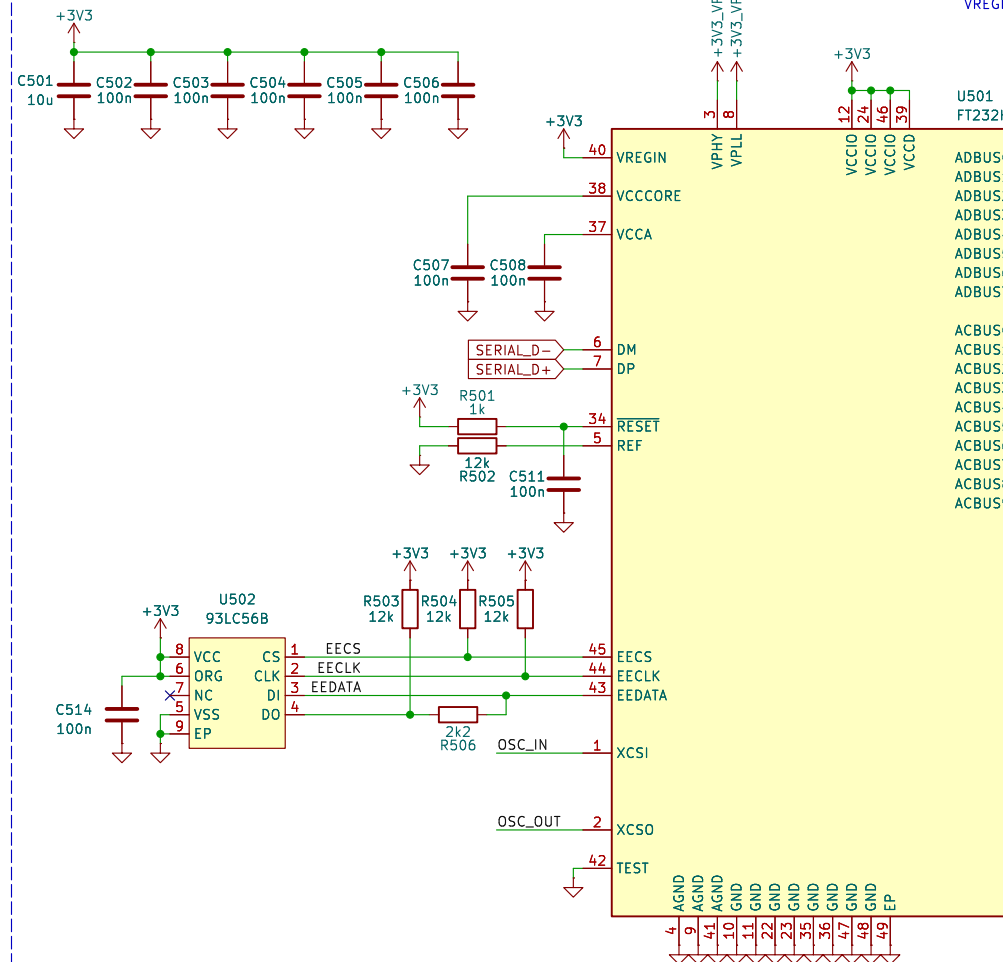


Ferrite beads must be rated at least for 600mA

Crystal



$(10\text{pF Load Capacitance} - 5\text{pF Stray Capacitance}) * 2 = 10\text{pF Load Capacitors}$



VREGIN = 5V --> VCCD = 3.3V OUTPUT
VREGIN = 3.3V --> VCCD = 3.3V INPUT

IO Banks

Bank	Pin	Signal
BANK 34	R2	IO_L1P_T0_34
	T2	IO_L1N_T0_34
	R3	IO_L2P_T0_34
	T3	IO_L2N_T0_34
	T1	IO_L3P_T0_DQS_34
	U1	IO_L3N_T0_DQS_34
	V2	IO_L5P_T0_34
	W2	IO_L5N_T0_34
	V3	IO_L6P_T0_34
	W3	IO_L6N_T0_VREF_34
	U3	IO_L9P_T1_DQS_34
	U2	IO_L9N_T1_DQS_34
	U4	IO_L11P_T1_SRCC_34
	W4	IO_L12N_T1_MRCC_34
	W5	IO_L12P_T1_MRCC_34
	V4	IO_L11N_T1_SRCC_34
	W7	IO_L13P_T2_MRCC_34
	W6	IO_L13N_T2_MRCC_34
	U8	IO_L14P_T2_SRCC_34
	V8	IO_L14N_T2_SRCC_34
	U5	IO_L16P_T2_34
	V5	IO_L16N_T2_34
	U7	IO_L19P_T3_34
	V7	IO_L19N_T3_VREF_34

RJ45 LEDs

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Sheet: /Serial/
File: serial.kicad_sch

Title: Serial

Size: A4
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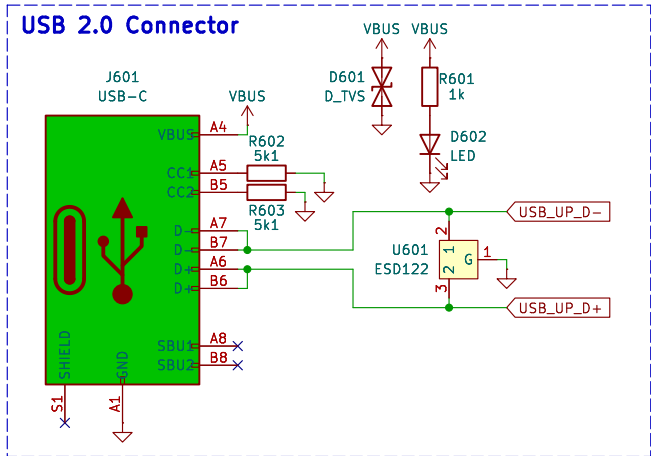
Date: 2026-03-11

Rev: 1.0

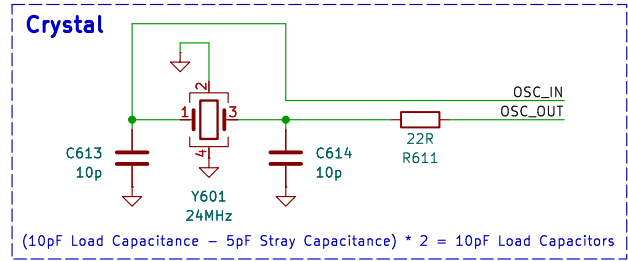
Id: 5/8

USB [6]

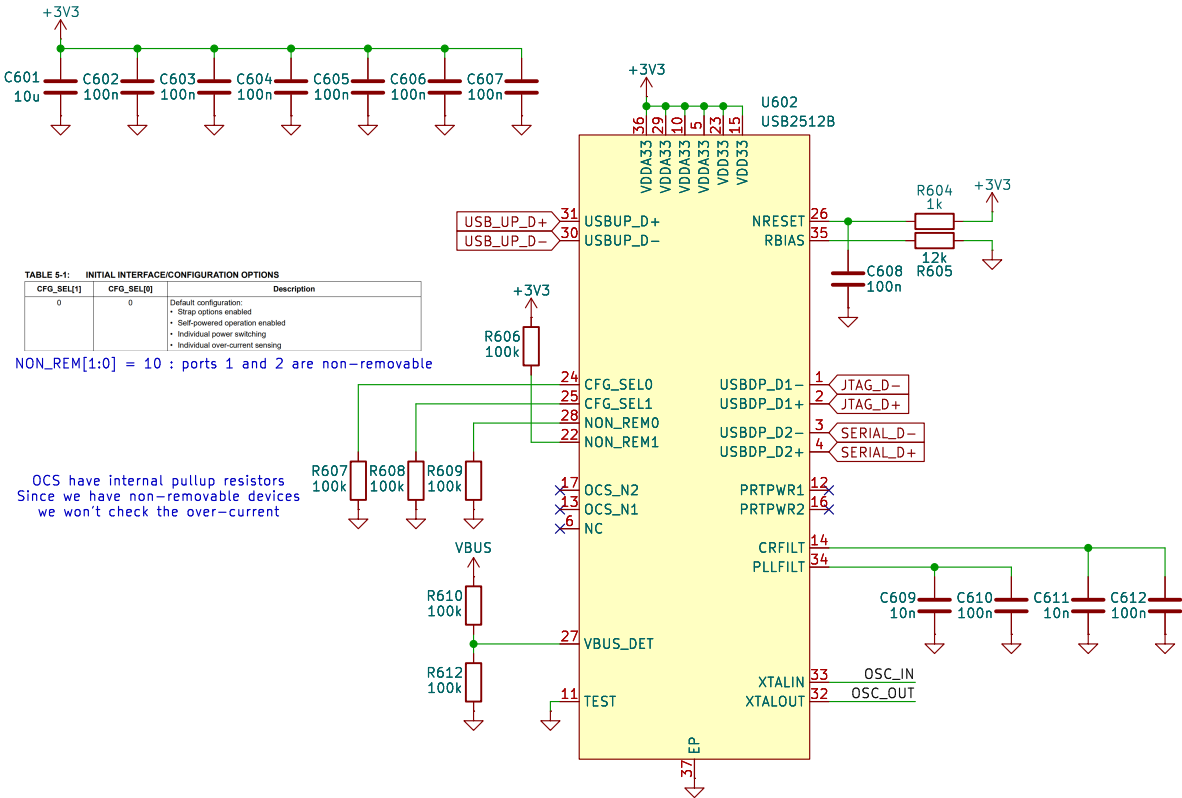
USB 2.0 Connector



Crystal



USB 2.0 Hub



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Sheet: /USB/
File: usb.kicad_sch

Title:

Size: A4
KiCad E.D.A. 9.0.6

Date:

Rev: 1.0
Id: 6/8

Title: Connectivity		
Size: A3	Date: 2026-03-12	Rev: 1.0
KiCad E.D.A. 9.0.6		Id: 7/8